

Computer-aided analysis of gas turbine cycles

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This paper describes a microcomputer-based, interactive, and menu-driven software package designed to help mechanical engineering students to understand gas turbines and to allow them to conduct more analysis of gas turbine cycles than they would normally be able to do by hand-calculation. The program deals with gas turbine cycle analysis so the acronym GTCA is used. GTCA is written in the Pascal computer language and runs on IBM PC, or compatible, computers. Improvements to the basic Brayton cycle, including three compressor and turbine stages, reheater, heat exchanger, intercooler, and precooler are incorporated into the program. The package is highly flexible and allows the user to model cycle schemes formed of any combination of these elements and to handle both shaft power turbines and aircraft turbojet and turbofan turbines. An important feature of the program is its ability to solve for any unknown variables. In addition to this, the program provides a schematic of the turbine plant layout and a temperature-entropy diagram of the cycle, and permits the plotting of the variation of any quantity versus any other quantity. This option enables the student to easily study and understand the effects of changing design variables on the overall performance of the cycle and permits its optimization. The statistical survey conducted along with the examples presented demonstrate the capabilities of the package as a teaching tool.

INTRODUCTION

Computer-aided design (CAD) packages are gaining in importance nowadays owing to the powerful computational and graphical capabilities existing in today's computers. As a result of this, the dependence on microcomputers in all fields of education is increasing. In particular, the introduction of microcomputers into engineering programs has relieved students of the lengthy computations required in solving complex processes. This in turn, has led students into a deeper understanding of the basic principles involved by allowing them to conduct more analysis of the problem than they would normally be able to do without a computer.

The basic gas turbine cycle is the Brayton cycle and it is usually introduced to mechanical engineering students in a course of engineering thermodynamics. The practical applications of the cycle to gas turbine plants for shaft power and aircraft propulsion along with additions to the cycle such as reheat, heat exchange, intercooling and precooling are usually covered in detail in a senior course which concentrates on the application of thermodynamics to gas turbines. Details of the subject matter can be found in various textbooks [1-4].

As a part of the course, students are expected to calculate the performance of several versions of the Brayton cycle used in generating electricity and powering aircraft. This is accomplished in a

step-by-step procedure starting with a basic Brayton cycle and progressively adding provision for reheat, intercooling, precooling and heat exchange. Owing to the heavy computations involved, few cases can usually be considered. Because of this, students are normally informed of the effects of varying any parameter on the overall performance of the cycle, but often do not have sufficient time to prove this for themselves, given the length of time required to calculate several examples by hand. With the widespread availability of microcomputers, the majority of engineering students now have access to these machines and, therefore, may exploit the state-of-the-art advances in this technology to solve time-consuming problems. To this end, the aim of the present paper is to present a program developed to provide the mechanical engineering student with a tool that allows the student to explore the effects of design changes on the performance of gas turbine plants without the boredom of performing hand-calculations.

DESCRIPTION OF THE PROGRAM

GTCA (gas turbine cycle analysis) is written in Pascal using the Turbo Pascal compiler version 6 [5] which includes a powerful debugger and facilitates enhanced graphics options. The package runs on any IBM PC, XT/AT or compatible which includes at least 640 Kbyte of main memory, a colour monitor equipped with a video or enhanced graphics adaptor card (VGA or EGA), and an optional maths coprocessor.

The software is divided into three major modules. The first one consists of procedures for generating menus and data entry windows. The role of the second module, which is composed of several procedures, is to solve for unknowns in different components of the cycle. An iterative solution procedure is adopted. In a given component of the cycle, solutions are obtained by iterating among unknowns. Iterations continue until no further change in the number of known variables is noticed. The program then steps to solve for unknowns in another component. The same procedure is repeated in all components. Once this is done, a global iteration is completed. The iterative process consists of repeating the solution all over again until no further changes in the number of known variables in any of the components occur. The purpose of the third module is to view, print, and plot results. The second and third modules are intercorrelated, owing to the need for repeatedly solving the whole cycle while plotting the variation of one quantity versus another. The program allows the plotting of the variation, as a function of a chosen variable, of up to three parameters at a time.

The menu structure of the program, depicted in Figs 1 and 2, is designed in such a way as to provide a user-friendly environment. The main menu offers three entries that allow the user to perform analysis of shaft power cycles or turbine cycles or to exit the program. By choosing shaft power cycles or turbine cycles the main menu disappears and a new menu of five entries is displayed. By selecting the appropriate task from this menu, the user can input data, solve for unknowns, optimize the cycle, or use the graphics screen as a window through which he/she can see and print a schematic of the cycle layout or of the temperature–entropy diagram. The user starts by choosing entry INPUT which allows him/her to enter the values of the known quantities. As depicted in Fig. 2, the menu-driven structure of the package facilitates this task by guiding the user throughout the data entering procedure. Having entered all data, the user may store them into a file that can be read by the program at any time. This option is included to permit easy correction of wrongly inputted or missing data. After the type of gas turbine cycle involved (shaft power cycle or gas turbine cycle for aircraft propulsion) has been chosen and all data has been entered, a solution is obtained by choosing the SOLVE entry. It should be mentioned that one can solve for a specific component in the cycle or for the whole cycle at once (Fig. 2). Results may be viewed and printed by going to the OUTPUT entry (Fig. 2). The effect of varying a parameter on the overall performance of the cycle or on the performance of an element in the cycle, is assessed as shown in Figs 1

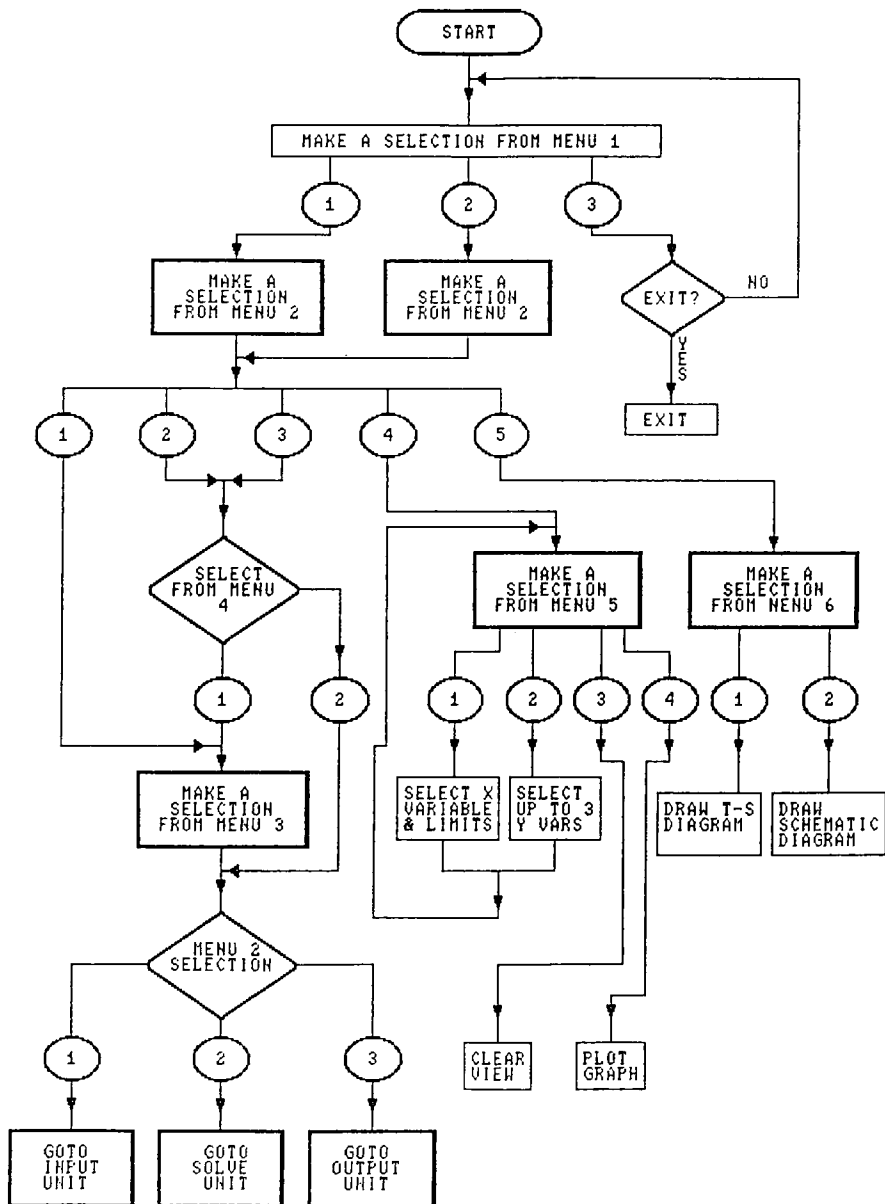


Fig. 1. Hierarchical menu structure of GTCA.

and 2, in the GRAPH entry. This allows a comprehensive analysis of the different parameters involved and permits the optimization of the cycle. A schematic and a temperature-entropy diagram of the cycle may be seen and manipulated from the DIAGRAM entry. The T-S diagram option is included to enable the user to visualize the cycle in these important thermodynamic

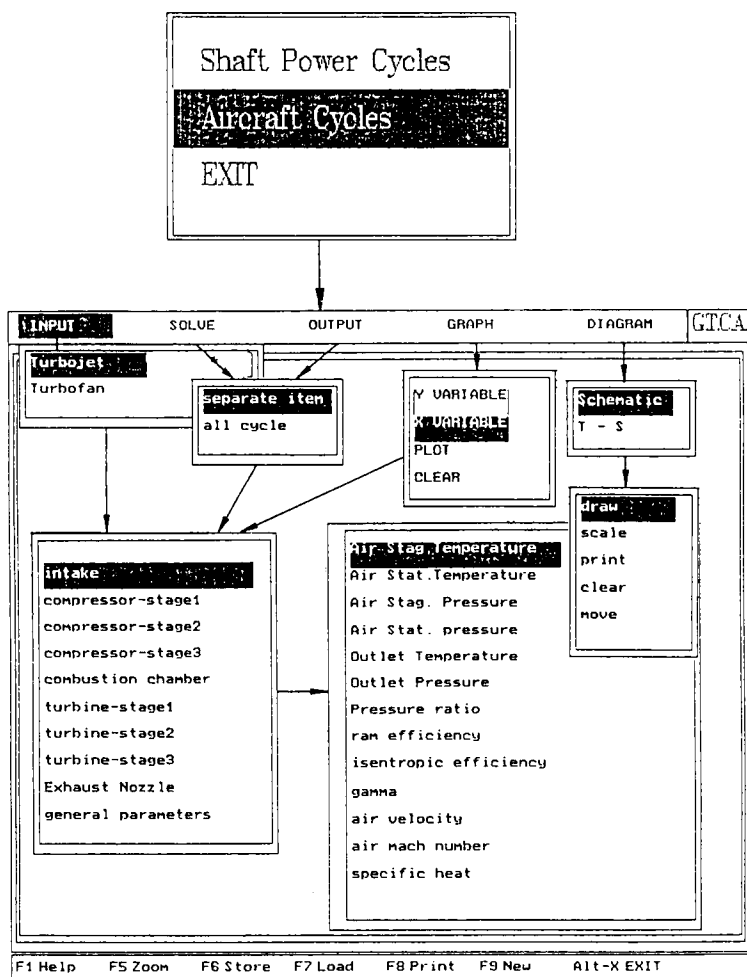


Fig. 2. Main menu and several illustrative submenus of GTCA.

coordinates. A scale command is included here to help magnify specific regions of the diagrams. To increase the usefulness of the package and to facilitate its usage, a **HELP** option, accessible from any level during execution, is included.

If the IBM 'graphics' module has been loaded at the beginning of the session, then a hard copy of both schematic and T-S diagrams can be obtained by using the 'print-screen' command with a suitable graphics printer.

EDUCATIONAL BENEFITS AND STUDENT ACCEPTANCE

This package can be used in parallel with a course on gas turbines to help students to enhance their understanding of the material. To test the usefulness of GTCA, the program was given to a total

number of 37 students attending this final-year course. Students were asked to use the program at their convenience because it was felt that providing a *variable pace environment* would allow best utilization of the package. As mentioned earlier, the full capabilities of the program permit easy evaluation of the performance of gas turbine cycles for shaft power and aircraft propulsion under different design conditions.

By the end of the term, a quantitative assessment of the package was performed. This involved presenting to all 37 users a questionnaire composed of the following six questions:

- (1) How user-friendly do you regard the package to be?
- (2) How do you rate the simplicity of the menu structure?
- (3) How effective is the HELP utility?
- (4) How effectively did the package aid your understanding of gas turbine cycles?
- (5) How do you rate the package as a teaching aid?
- (6) How effective is the graphics representation and plots?

In answering the above questions, students had to choose one of the following: 'excellent', 'very good', 'good', or 'fair'. The results of the questionnaire are shown in Table 1, from which one may conclude that the package proved to be very beneficial as a teaching aid.

As can be seen from Table 1, the overall evaluation of the package according to the above six questions, was rated as 'excellent' by 62% of the students and as 'very good' by 21% of the students.

Table 1. Statistics showing students' evaluation results

Question number	Evaluation			
	Excellent	Very good	Good	Fair
(1)	21	8	5	3
(2)	26	5	4	2
(3)	19	7	7	4
(4)	22	9	3	3
(5)	20	10	4	3
(6)	29	7	1	0
Percentage overall rating	61.72	20.72	10.81	6.75

REPRESENTATIVE PROBLEMS

The capabilities of the program are demonstrated in this section by presenting the solutions to three test problems. The first problem deals with a shaft power cycle, while the second and third problems are applications to turbojet and turboprop cycles, respectively. Records of the input and output data for problems 1, 2, and 3 are presented in Tables 2, 3, and 4, respectively.

The schematic diagrams of the cycles considered are shown in Fig. 3. As depicted in Fig. 3(a), the shaft power cycle scheme (Problem 1) comprises two compressor and two turbine stages, an intercooler, a heat exchanger, a combustion chamber, and a reheater. The schematic diagrams for

Input data

G.T.C.A.
Gas Turbine Cycle Analysis
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```

=====
COMPRESSOR STAGE - 1
=====
Inlet Temperature : 286.15(K)
Inlet Pressure : 1.000
Pressure Ratio : 2.500
gamma : 1.400
Isentropic eff : 0.850
=====
INTERCOOLER
=====
Outlet Temperature : 300.000
Pressure Drop : 0.048
gamma : 1.400
=====
COMPRESSOR STAGE - 2
=====
Inlet Temperature : 300.000
Pressure Ratio : 2.500
gamma : 1.400
Isentropic eff : 0.850
=====
COMBUSTION CHAMBER
=====
Outlet Temperature : 1000.000
Pressure Drop : 0.000
Inlet Gamma : 1.400
Outlet Gamma : 1.333
Combustion Efficiency: 1.000
=====
TURBINE STAGE - 1
=====
Inlet Temperature : 1000.000
gamma : 1.333
Isentropic eff : 0.850
=====
COMBUSTION CHAMBER
=====
Outlet Temperature : 1000.000
Pressure Drop : 0.126
Inlet Gamma : 1.333
Outlet Gamma : 1.333
Combustion Efficiency: 1.000
=====
TURBINE STAGE - 2
=====
Inlet Temperature : 1000.000
gamma : 1.333
Isentropic eff : 0.850
=====
HEAT EXCHANGER
=====

```

```

=====
COMPRESSOR STAGE - 2
=====
Inlet Temperature : 300.000
Inlet Enthalpy : 301.350
Inlet Pressure : 2.507
Outlet Temperature : 405.422
Outlet Enthalpy : 405.148
Outlet Pressure : 2.547
Pressure Ratio : 106.098
gamma : 1.400
polytropic eff : 0.868
Isentropic eff : 0.950
temperature ratio : 1.352
specific heat : 1.005
=====
COMBUSTION CHAMBER
=====
Inlet Temperature : 787.118
Inlet Pressure : 289.640
Inlet Enthalpy : 1000.000
Outlet Temperature : 5.222
Outlet Pressure : 1148.862
Outlet Enthalpy : 212.382
Temperature Diff. : 0.000
Pressure Drop : 1.005
Inlet Specific Heat : 1.400
Outlet Specific Heat : 1.333
Inlet Gamma : 1.400
Outlet Gamma : 1.333
Isentropic Eff : 0.850
Theoretical F/A Ratio: 0.307
Combustion Efficiency: 1.000
% Pressure Drop : -0.000
=====
TURBINE STAGE - 1
=====
Inlet Temperature : 1000.000
Inlet Enthalpy : 1148.862
Inlet Pressure : 5.992
Outlet Temperature : 816.496
Outlet Enthalpy : 212.618
Outlet Pressure : 2.568
Pressure Ratio : 207.244
gamma : 1.333
polytropic eff : 0.854
Isentropic eff : 0.950
temperature ratio : 1.352
specific heat : 1.149
=====
COMBUSTION CHAMBER
=====
Inlet Temperature : 619.409
Inlet Pressure : 2.506

```

```

=====
Pressure Drop : 0.275
gamma : 1.400
=====
HOT SIDE
=====
Outlet Pressure : 1.070
Outlet Enthalpy : 1.222
gamma : 1.400
M.E. Effectiveness : 0.800
=====
GENERAL PARAMETERS
=====
mechanical efficiency : 1.000
Work Generated : 122.000

```

Output data

```

=====
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=====

```

COMPRESSOR STAGE - 1

```

=====
Inlet Temperature : 286.000
Inlet Enthalpy : 287.237
Inlet Pressure : 1.000
Outlet Temperature : 286.972
Outlet Enthalpy : 287.237
Outlet Pressure : 2.500
Pressure Ratio : 2.500
Work : -101.146
gamma : 1.400
polytropic eff : 0.868
isentropic eff : 0.950
Temperature ratio : 1.200
specific heat : 1.005
=====

```

INTERCOOLER

```

=====
Inlet Temperature : 286.972
Inlet Enthalpy : 287.237
Inlet Pressure : 300.000
Outlet Temperature : 301.350
Outlet Enthalpy : 301.350
Outlet Pressure : 2.507
Pressure Drop : 0.068
Heat extracted : 87.082
gamma : 1.400
Temperature Drop : 84.378
Specific Heat : 1.005
=====

```

HEAT EXCHANGER

```

=====

```

COLD SIDE

```

=====
Inlet Temperature : 405.622
Inlet Pressure : 4.267
Inlet Enthalpy : 426.118
Outlet Temperature : 787.118
Outlet Pressure : 2.992
Outlet Enthalpy : 787.660
Temperature Diff. : 281.496
Pressure Drop : 0.275
Specific Heat : 1.005
gamma : 1.400
=====

```

HOT SIDE

```

=====
Inlet Temperature : 882.492
Inlet Pressure : 1.196
Inlet Enthalpy : 1012.862
Outlet Temperature : 948.024
Outlet Enthalpy : 1012.862
Outlet Pressure : 2.507
Temperature Diff. : 332.539
Pressure Drop : 0.166
Heat extracted : 1.149
Specific Heat : 1.005
gamma : 1.400
=====

```

M.E. Effectiveness

```

=====
Heat Exchanged : 287.212
gamma : 1.400
=====

```

GENERAL PARAMETERS

```

=====
mechanical efficiency : 1.000
cycle efficiency : 0.222
isentropic eff : 0.977
total F/A ratio : 0.014
Work Generated : 122.000
=====

```

Table 2. Input and output data for a shaft power cycle (Problem 1)

Input data

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```

*****
INTEAKE
*****
Air Stat. Temperature : 291.987
Air Stat. Pressure : 0.540
Isentropic efficiency : 0.920
gamma : 1.400
Air velocity : 270.000

COMPRESSOR STAGE - 1
*****
Pressure Ratio : 8.000
gamma : 1.400
Isentropic eff : 0.870

COMBUSTION CHAMBER
*****
Outlet Temperature : 1200.000
Inlet Gamma : 1.400
Outlet Gamma : 1.333
Combustion Efficiency : 0.980
Pressure Drop : 0.040

TURBINE STAGE - 1
*****
Inlet Temperature : 1200.000
gamma : 1.333
Isentropic eff : 0.900

HOT NOZZLE
*****
Ambient Pressure : 0.540
Isentropic Efficiency : 0.950
Gamma : 1.333
Nozzle Status :

GENERAL PARAMETERS
*****
C.F.V. : 42100.00
Mechanical Efficiency : 0.990
Cruising Speed : 270.000
  
```

Output data

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```

*****
INTEAKE
*****
Air Stat. Temperature : 291.987
Air Stat. Pressure : 0.540
Isentropic efficiency : 0.920
gamma : 1.400
Air velocity : 270.000
Mach number : 0.842
Specific heat : 1.005

COMPRESSOR STAGE - 1
*****
Inlet Temperature : 291.987
Inlet Enthalpy : 279.301
Inlet Pressure : 0.540
Outlet Temperature : 544.222
Outlet Enthalpy : 566.862
Outlet Pressure : 6.673
Pressure Ratio : 8.000
Work : -273.561
gamma : 1.400
polytropic eff : 0.902
Isentropic eff : 0.870
Temperature ratio : 1.923
Specific Heat : 1.005

COMBUSTION CHAMBER
*****
Inlet Temperature : 544.222
Inlet Pressure : 6.673
Inlet Enthalpy : 566.862
Outlet Temperature : 1200.000
Outlet Pressure : 1378.634
Outlet Enthalpy : 635.678
Temperature Diff. : 0.267
Pressure Drop : 1.004
Inlet Specific Heat : 1.400
Inlet Gamma : 1.400
Outlet Specific Heat : 1.149
  
```

```

Outlet Gamma      | 1.332
Actual F/A Ratio- | 0.018
Theoretical F/A Ratio | 0.018
Combustion Efficiency | 0.980
% Pressure Drop    | 0.040

=====
TURBINE STAGE - 1
=====
Inlet Temperature | 1294.000
Inlet Enthalpy    | 1278.624
Inlet Pressure    | 6.406
Outlet Temperature | 959.480
Outlet Enthalpy   | 1152.310
Outlet Pressure   | 1.231
Pressure Ratio    | 5.224
Gamma            | 1.332
polytropic eff   | 0.888
isentropic eff    | 0.990
temperature ratio | 1.251
specific heat     | 1.149

=====
HOT NOZZLE
=====
Ambient Pressure | 0.540
Inlet Temperature | 959.480
Inlet Pressure    | 2.127
Outlet Stat. Temperature | 822.529
Outlet Stat. Pressure | 1.218
Outlet Density    | 0.216
Jet Exit Velocity | 560.760
Critical Pressure | 1.185
Critical Temperature | 822.529
Critical Speed of Sound | 560.960
Isentropic Efficiency | 0.950
Exit Area/Unit Mass | 0.002
Gamma            | 1.332
Specific Heat     | 1.149
Nozzle Status    | choked

=====
GENERAL PARAMETERS
=====
Specific Thrust   | 525.004
Propulsive Efficiency | 0.770
Energy Conversion Eff. | 0.150
Overall Efficiency | 0.176
C.F.V.           | 43100.00
S.F.C.           | 0.128
Mechanical Efficiency | 0.990
Specific Total Tur. Work | 276.324
Specific Total Comp Work | 4255.361
Total Combs. Pr. Ratio | 6.000
Total Turb. Pr. Ratio | 2.741
Cruising Speed    | 270.000

```

Table 3. Input and output data for a turbojet cycle (Problem 2)

Input data

G.T.C.A.	
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INLET E	
Condition	
Air Stat. Temperature	1 225.700
Air Stat. Pressure	1 0.540
Isentropic efficiency	1 1.000
gamma	1 1.400
Air velocity	1 270.000

COMPRESSOR STAGE - 1	

Pressure Ratio	1 1.720
gamma	1 1.400
Isentropic eff	1 0.880

COMPRESSOR STAGE - 2	

gamma	1 1.400
Isentropic eff	1 0.800

COMBUSTION CHAMBER	

Outlet Temperature	1 1400.000
Inlet Gamma	1 1.400
Outlet Gamma	1 1.332
Combustion Efficiency	1 0.990
% Pressure Drop	1 0.025

TURBINE STAGE - 1	

Inlet Temperature	1 1400.000
gamma	1 1.332
Isentropic eff	1 0.880

TURBINE STAGE - 2	

gamma	1 1.332
Isentropic eff	1 0.880

HOT NOZZLE	

Ambient Pressure	1 0.540
Isentropic Efficiency	1 0.960
Gamma	1 1.332
Nozzle Status	

COLD NOZZLE	

Ambient Pressure	1 0.540
Isentropic Efficiency	1 0.960
Gamma	1 1.400
Nozzle Status	

COMPRESSOR STAGE - 2	

Inlet Temperature	1 347.597
Inlet Enthalpy	1 349.141
Inlet Pressure	1 1.479
Outlet Temperature	1 712.675
Outlet Enthalpy	1 151.862
Outlet Pressure	1 1.720
Pressure Ratio	1 9.884
Work	1 -266.721
gamma	1 1.400
polytropic eff	1 0.912
Isentropic eff	1 0.880
temperature ratio	1 2.000
specific heat	1 1.002

COMBUSTION CHAMBER	

Inlet Temperature	1 712.675
Inlet Pressure	1 14.320
Inlet Enthalpy	1 712.882
Outlet Temperature	1 1400.000
Outlet Enthalpy	1 1408.407
Temperature Diff.	1 687.322
Pressure Drop	1 0.512
Inlet Specific Heat	1 1.002
Inlet Gamma	1 1.400
Outlet Specific Heat	1 1.332
Outlet Gamma	1 0.921
Isentropic F/A Ratio	1 0.924
Theoretical F/A Ratio	1 0.924
Combustion Efficiency	1 0.990
% Pressure Drop	1 0.025

TURBINE STAGE - 1	

Inlet Temperature	1 1400.000
Inlet Enthalpy	1 1408.407
Inlet Pressure	1 14.109
Outlet Temperature	1 1074.282
Outlet Enthalpy	1 1234.202
Outlet Pressure	1 4.128
Pressure Ratio	1 3.418
Work	1 274.205
gamma	1 1.332
polytropic eff	1 0.862
Isentropic eff	1 0.880
temperature ratio	1 1.202
specific heat	1 1.149

TURBINE STAGE - 2	

Inlet Temperature	1 1074.282
Inlet Enthalpy	1 1234.202
Inlet Pressure	1 4.128
Outlet Temperature	1 852.401
Outlet Enthalpy	1 994.801
Outlet Pressure	1 1.523
Pressure Ratio	1 2.709
Work	1 229.401
gamma	1 1.332
polytropic eff	1 0.866
Isentropic eff	1 0.880
temperature ratio	1 1.241

GENERAL PARAMETERS

C.F.V. : 42100.00
 Air Mass Flow Rate : 95.000
 Mechanical Efficiency : 0.980
 Total Comp. Pr. Ratio : 17.000
 Cruising Speed : 270.000
 Bypass Ratio : 5.200

Output data

G.T.C-3.
 S4s Turbine Cycle Analysis
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INTAKE

Air Std. Temperature : 291.987
 Air Std. Pressure : 252.700
 Air Std. Density : 0.860
 Air Std. Enthalpy : 94.000
 Air Std. Velocity : 291.987
 Outlet Pressure : 0.860
 Pressure Ratio : 1.591
 Ram efficiency : 1.000
 Isentropic efficiency : 1.000
 gamma : 1.400
 Air velocity : 270.000
 Mach number : 0.842
 Specific heat : 1.005

COMPRESSOR STAGE - 1

Inlet Temperature : 291.987
 Inlet Enthalpy : 94.000
 Inlet Pressure : 0.860
 Inlet Density : 0.860
 Inlet Temperature : 247.297
 Outlet Enthalpy : 349.161
 Outlet Pressure : 1.479
 Pressure Ratio : 1.720
 Work : 152.860
 gamma : 1.400
 Isentropic eff : 0.889
 Isentropic eff : 0.889
 temperature ratio : 1.190
 specific heat : 1.005

specific heat : 1.149
 HOT NOZZLE
 Mass Flow Rate : 22.619
 Ambient Pressure : 0.240
 Inlet Temperature : 865.901
 Inlet Pressure : 1.522
 Inlet Density : 742.207
 Outlet Stat. Temperature : 0.800
 Outlet Stat. Pressure : 0.112
 Jet Exit Area : 522.902
 Outlet Density : 0.800
 Jet Exit Velocity : 0.800
 Critical Temperature : 742.207
 Critical Pressure : 522.902
 Isentropic Efficiency : 0.960
 Exit Area/Unit Mass : 0.002
 gamma : 1.332
 Specific Heat : 1.149
 Nozzle Status : choked

COLD NOZZLE

Mass Flow Rate : 72.281
 Ambient Pressure : 0.240
 Inlet Temperature : 247.297
 Inlet Pressure : 1.479
 Inlet Density : 742.207
 Outlet Stat. Temperature : 0.799
 Outlet Stat. Pressure : 0.112
 Jet Exit Area : 522.902
 Outlet Density : 0.800
 Jet Exit Velocity : 0.800
 Critical Temperature : 742.207
 Critical Pressure : 522.902
 Isentropic Efficiency : 0.960
 Exit Area/Unit Mass : 0.002
 gamma : 1.400
 Specific Heat : 1.005
 Nozzle Status : choked

GENERAL PARAMETERS

Momentum Thrust : 1106.85
 Pressure Thrust : 8007.210
 Net Thrust : 19104.20
 Specific Thrust : 201.097
 Propulsive Efficiency : 0.852
 Overall Efficiency : 0.258
 C.F.V. : 42100.00
 Fuel mass Flow Rate : 0.464
 Air mass Flow Rate : 95.000
 S.F.C. : 0.087
 Mechanical Efficiency : 1379.17
 Total Compressor Work : 12601.5
 Specific Total Tur. Work : 146.097
 Specific Total Comp Work : 142.175
 Total Turbine Pr. Ratio : 17.000
 Total Turbine Pr. Ratio : 17.000
 Cruising Speed : 270.000
 Bypass Ratio : 5.200
 Hot Mass Flowrate : 72.281
 Bypass Ratio : 3.200

Table 4. Input and output data for a turbofan cycle (Problem 3)

the turbojet (Problem 2) and turbofan (Problem 3) cycles are presented in Figs 3(b) and 3(c). Both cycles contain two compressor and turbine stages, an intake, a combustion chamber and a nozzle. In the turbofan cycle, part of the air compressed by the fan (or the low-pressure compressor) is directly expanded in a cold nozzle while the rest proceeds through the engine and expands in the hot nozzle.

In addition to solving for unknown variables, the main educational benefits of the package are realized when studying the effects of parameters on the overall performance of the cycle. Examples of such studies are depicted in Figs 4–9. Figs 4(a) and 4(b) show the effects of varying the reheater outlet temperature of the shaft power cycle (Problem 1) on the overall cycle efficiency and the fuel/air ratio, respectively. These plots may be generated, depending on the speed of the PC used and the availability of a maths coprocessor, in less than 1 minute (or a few seconds with an advanced PC). This is accomplished by choosing the GRAPH option and then selecting the x and y variables (up to three at a time). After entering the range of variation of x and pressing the 'plot' key, the program subdivides the x interval into 100 sub-intervals and solves the whole cycle 100 times in order to compute and store the y values at the respective x values. Then, as shown, the y s are plotted

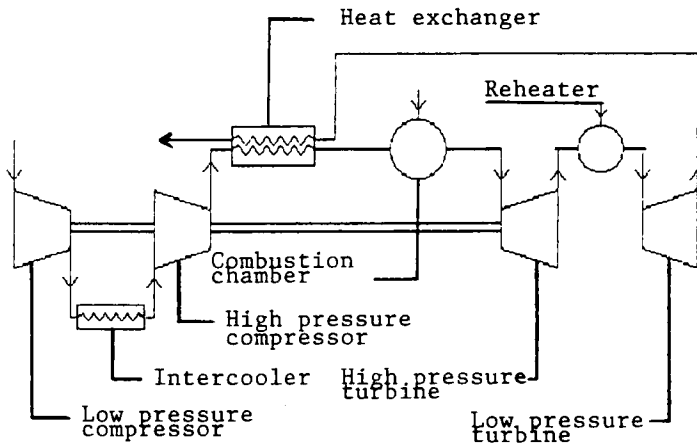


Fig. 3(a). Schematic diagram for a shaft power cycle (Problem 1).

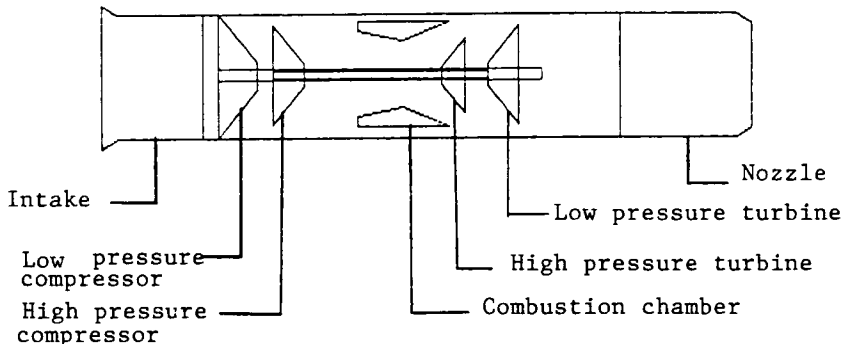


Fig. 3(b). Schematic diagram for a turbojet cycle (Problem 2).

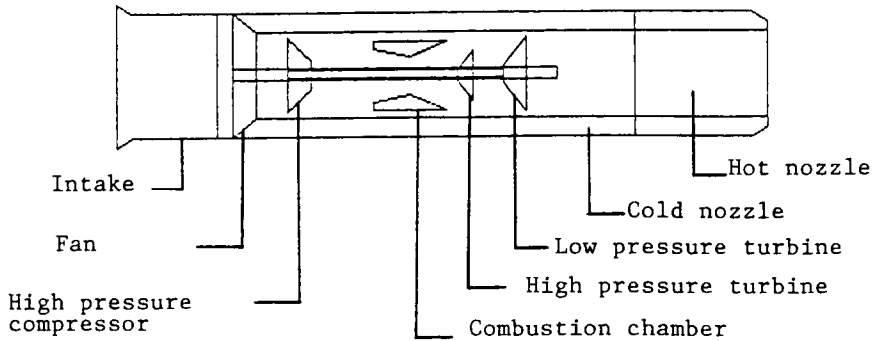


Fig. 3(c). Schematic diagram for a turbofan cycle (Problem 3).

as functions of x . From the relations in Figs 4(a) and 4(b), it is clear that, at constant output work, increasing the reheater outlet temperature decreases the cycle efficiency and increases the fuel/air ratio. This is to be expected since more fuel is added to accomplish the same task. The student may experiment more by changing other parameters and exploring their effects on the cycle performance. The T-S diagram of the shaft power cycle is presented in Fig. 5. The irreversibility of the compression and expansion processes are demonstrated by their inclination towards higher entropy. This plot also shows the heat addition in the combustion chamber (5–6), heat addition in the

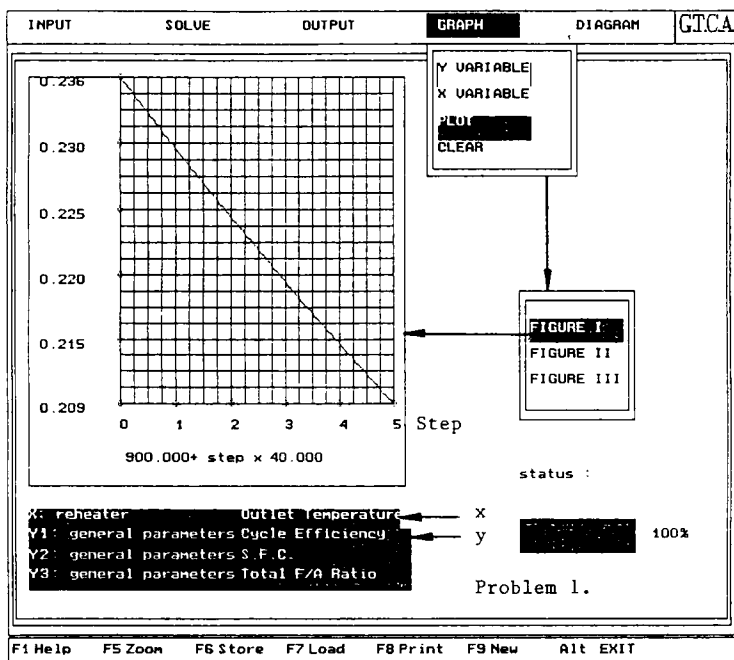


Fig. 4(a). Variation of cycle efficiency with reheater outlet temperature.

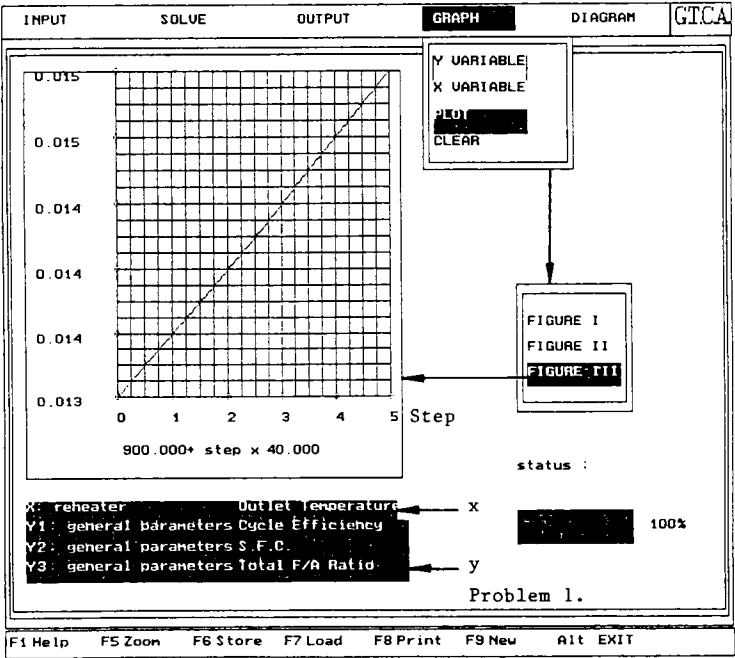


Fig. 4(b). Variation of fuel/air ratio with reheat outlet temperature.

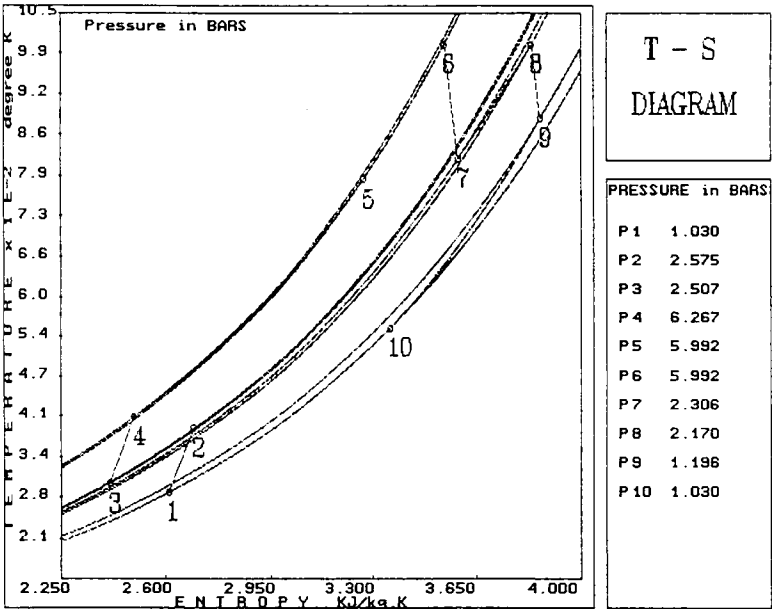


Fig. 5. T-S diagram for a shaft power cycle (Problem 1).

reheater (7–8), heat lost (10–1) and heat exchanged (4–5) processes. Owing to friction, these processes do not occur at constant pressures and the pressure drops are clearly seen on the T–S diagram.

Figs 6(a) and 6(b) reveal the effects of the intake isentropic efficiency on the net specific thrust and the overall efficiency of the turbojet cycle (Problem 2). These two quantities are found to increase with increasing intake isentropic efficiency. These physically plausible results are to be expected since with increasing the intake isentropic efficiency, the inlet stagnation pressure to the first compressor will be higher and less work will be needed for the pressure to be raised to its desired level. This, in turn, implies less work to be transmitted from the turbine to the compressors and consequently an increase in the overall efficiency and the specific thrust of the cycle. The T–S diagram of the cycle is shown in Fig. 7. All processes are presented using stagnation temperature and pressure. Using the scale option the process (5–6) is magnified. This process is expansion, at constant stagnation temperature, of the gases in the nozzle.

For the turbofan cycle (Problem 3), the effects of varying the fan pressure ratio on the net specific thrust and cycle efficiency are shown in Figs 8(a) and 8(b), respectively. From these plots, the student learns that there is an optimal value for the fan pressure ratio which maximizes the net specific thrust and cycle efficiency and as such this ratio should be the one used in design. The cycle T–S diagram along with magnified regions of it are shown in Fig. 9.

As demonstrated by the above examples the package is educationally very beneficial. It relieves the student from the tedious calculations needed in solving gas turbine problems and, most importantly, it gives him/her an opportunity to easily study the effects of design variable variations on cycle performance.

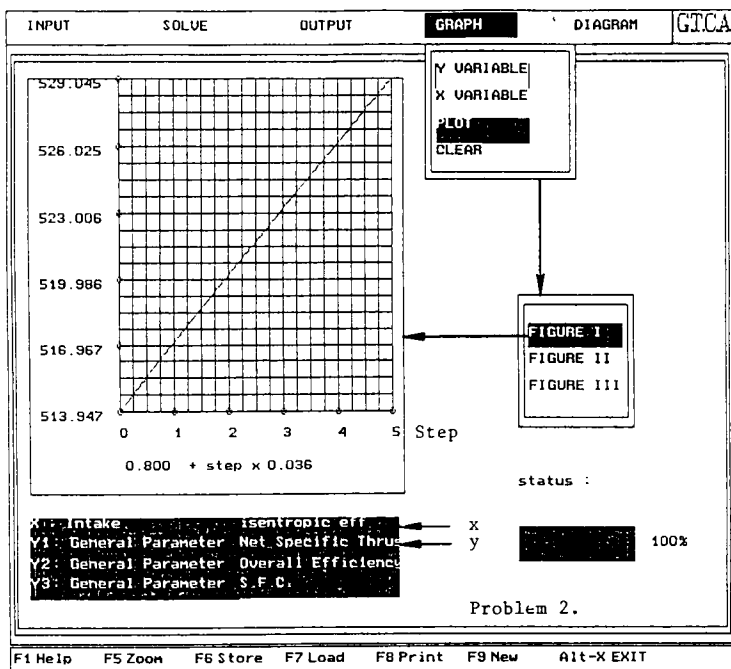


Fig. 6(a). Variation of net specific thrust with intake isentropic efficiency.

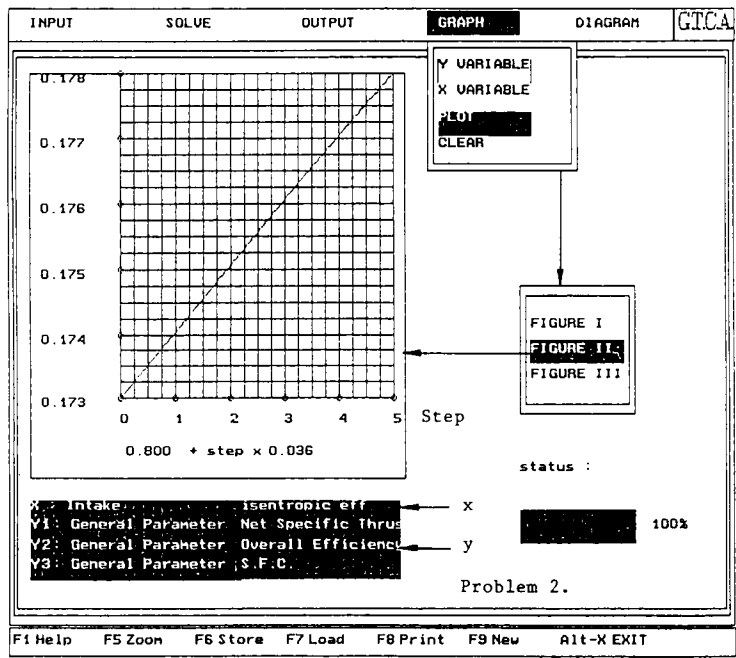


Fig. 6(b). Variation of overall cycle efficiency with intake isentropic efficiency.

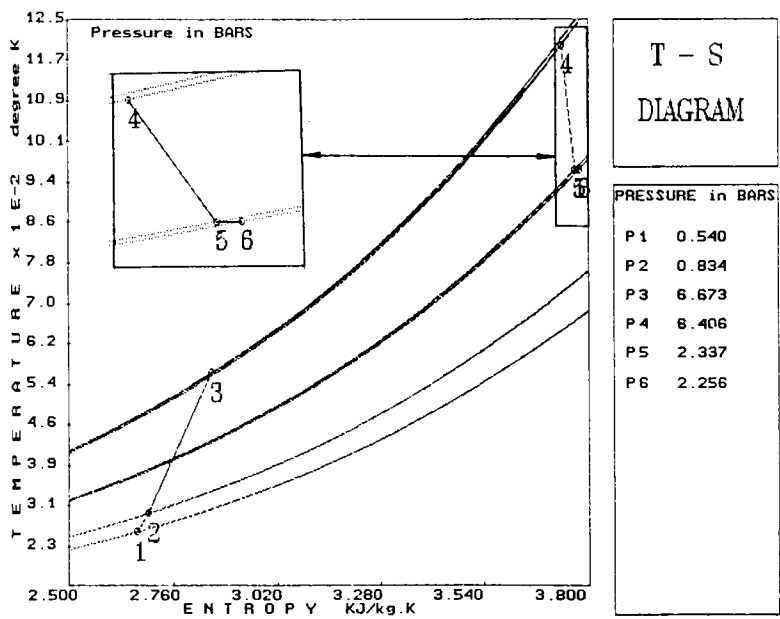


Fig. 7. T-S diagram for a turbojet cycle (Problem 2).

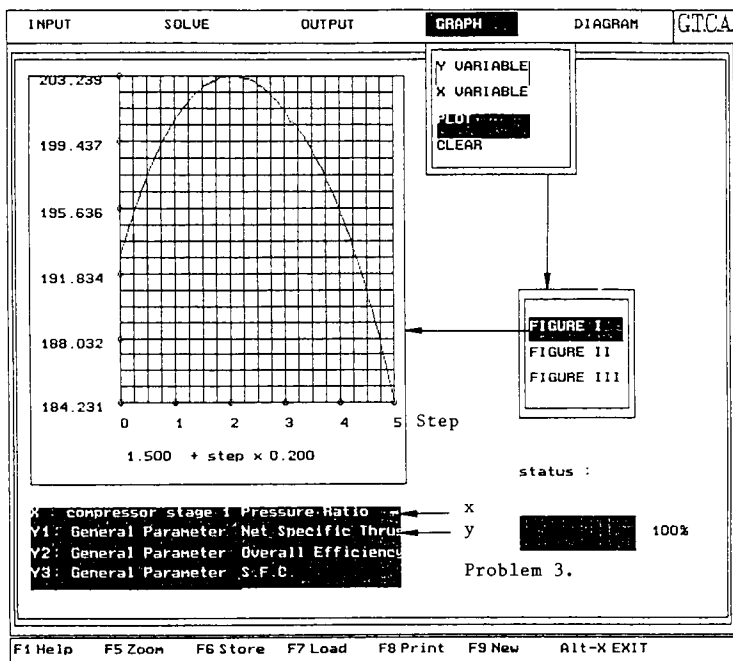


Fig. 8(a). Variation of net specific thrust with fan pressure ratio.

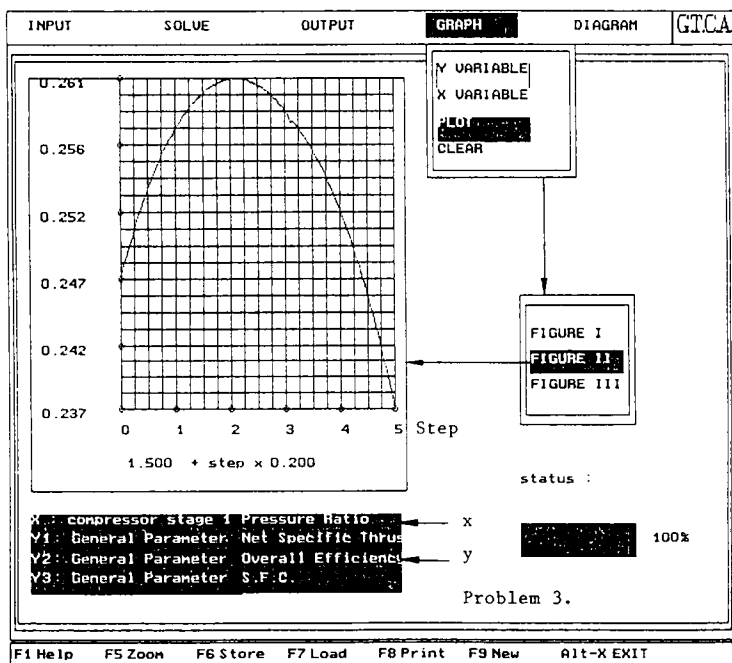


Fig. 8(b). Variation of cycle efficiency with fan pressure ratio.

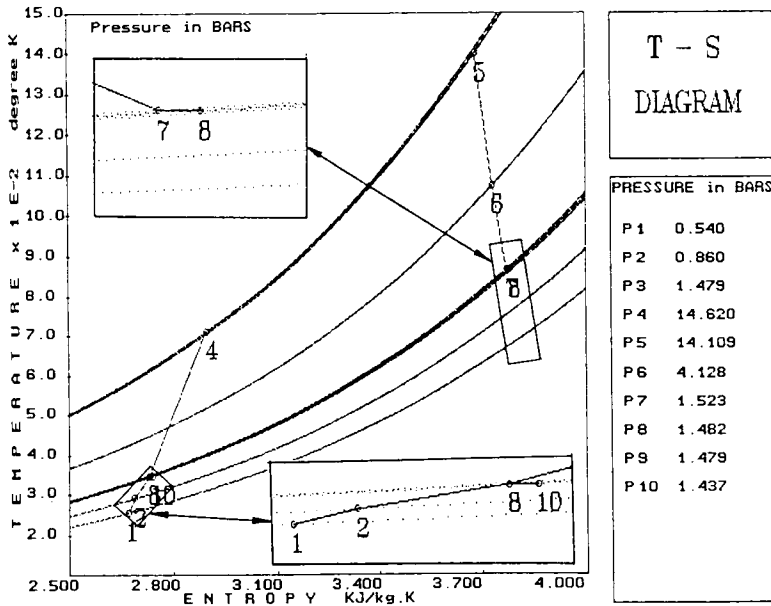


Fig. 9. T-S diagram for a turbofan cycle (Problem 3).

CONCLUSIONS

A microcomputer-based program for analysis of gas turbine cycles for both shaft power and aircraft propulsion has been developed. The menu-driven structure of the program along with its help utility and enhanced graphical capabilities allow the mechanical engineering student to easily explore the effect of changing design parameters on cycle performance. The package also provides the user with an efficient educational tool on a low-cost workstation. The statistical survey conducted shows the enthusiasm with which the program is received by students. The capabilities of the package as a teaching tool were demonstrated by presenting three illustrative example problems. The range of applications of the package is not confined to the configurations stated in the paper because, when this educational tool is combined with hand-calculations, the user will be able to solve a wider range of problems. Copies of the program will be provided to users on request to the authors.

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